Introduction To Applied Animal Breeding

Introduction Cont…

- **Animal breeding** is the application of genetic principles to genetically improve the efficiency of production of farm animals, particularly all the food-producing species

- **Genetics** provides the **foundation principles** which are utilized and synthesized into **breeding programs**

- The **tools** of genetic engineering, such as **marker-assisted selection** and transgenics are enhancing the work of animal breeders, enabling genetic progress to occur much more rapidly

Introduction Cont…

- It is the **careful selection** and **controlled assortative mating** of animals with the intention to get progeny whose average performance level is closer to the defined breeding goal (objective) than the average performance level of the parents‘ generation

- Requirement for successful (applied animal) breeding is the knowledge of:
  – The principles of **selection** and **mating systems,** and
  – Their implications on allele frequencies and genetic variation in a population

Introduction Cont…

– Breeding **without a breeding goal** is not feasible!

• Breeding goals directs "**Where to go**"

– Breeding means **coordination** of the several **breeding measures (steps)** and connected breeding actions

• **How to get there ??**

– **Designing suitable breeding program**

* **NB: Animal breeding is different from mere reproduction and not reproductive physiology**

- It is the study of how genes: Function, Inherited and Interact

Major tasks in animal breeding

- **Selection** of the most desirable animals to be parents of the next generation based on **predicted genetic merit**, i.e. **worth of an animal as a parent**
- **Production of superior genotypes** by the combination of genetics through breeding plans and systems of mating
- **Selection is the process that determines**:
  – Which individuals (animals) become parents,
  – How many offspring they produce, and
  – How long they remain in the population
- Selection **does not create new genes**,
- But by permitting animals possessing certain genes to leave more offspring it increases the frequency of the more desirable alleles

Selection is of two types:

- **Natural selection, or survival of the fittest (Darwin's theory):** the animals best adapted to their environment survive and produce the largest number of offspring. This is *natural selection, or survival of the fittest.*
- Variations produced by **mutation and recombination** of genetic factors, eliminates the unsuccessful genetic combinations and allows the most successful to multiply.
- Animal Breeders cannot ignore natural selection because natural selection affects wild and domestic species
- Example, all animals with lethal genetic defects are naturally selected against – they never live to become parents

Artificial selection . . .

- Selection that is **under human control**

  – It is the method used by breeders to make long-term genetic change in animals

  Artificial selection **has 2 aspects:**

  – **Replacement selection**: The process that determines which animals will become parents for the first time;

  – **Culling:** The process that determines which parents will no longer remain parents

Effectiveness of selection

- The **initial task** is to obtain an accurate estimate of the **additive BVs** of the animals available for selection
- Predicted from available information, such as their **own performance** and their **relative's performance**
- The *effectiveness or rate of improvement* in performance from selection during a generation is dependent on **3 primary factors**:

- *accuracy of selection,*
- *intensity of selection and*
- *generation interval*

$(\bar{p}_s - \bar{p})$

Effectiveness of selection . . .

$$\Delta G = h^2 (\bar{p}_s - \bar{p})$$

Response to selection (R) = Genetic Improvement (ΔG)

where

- ΔG is expected genetic change as a result of selection,
- h^2 is heritability of the trait,
- $(\bar{p}_s - \bar{p})$ is selection differential (= the difference b/n the mean of the selected parents Ps and the mean of the population P)

Effectiveness of selection . . .

- The **expected genetic improvement per year** is an important consideration

- ***Genetic improvement per year*** =

$$\frac{\text{accuracy} \times \text{intensity} \times \text{genetic variation}}{\text{years per generation (or selection cycle)}}$$

- **Example 1**

We wish to improve **ADG** in pigs.
$h^2 = 0.4$
herd mean ($\bar{p}$) = **1.6** lb/day
mean of individuals selected to be parents ($\bar{p}_s$) = **1.8** lb/day

selection differential (SD) = 1.8 - 1.6 = 0.2 lb/day

Response (R) = h^2 x SD = 0.4 x 0.2 = **0.08** lb/day per generation

new herd average after one generation of selection
1.6 + .08 = **1.68 lb/day**

Effectiveness of selection . . .

This simple formula assumes ***equal selection pressure in both sexes.***

This is, of course, seldom the case. More pressure can usually be applied in the **males** b/c ***fewer males are needed as replacements.***

The **SD** is then simply the average of the male & female selection differentials: $\mathbf{SD = (SD_M + SD_F) / 2}$.

Effectiveness of selection . . .

Example 2: We wish to decrease backfat thickness in pigs.

$h^2 = 0.5$

mean of all boars ($\bar{p}$) = 1.0 in; ***mean of boars selected*** ($\bar{p}_s$) to be replacements = 0.8 in

mean of all gilts ($\bar{p}$) = 1.2 in; ***mean of gilts selected*** ($\bar{p}_s$) to be replacements = 1.1 in

overall herd mean = 1.1 in

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selection differential (**SD**):

males	0.8 - 1.0 = -0.2 in
females	1.1 - 1.2 = $\bar{p}_s$1 in
overall	(-0.2 + -0.1) / 2 = -0.15 in

Response

$R = h^2 \times SD = 0.5 \times -0.15 = -.075$ in per generation

new herd mean = 1.1 + -0.075 = 1.025 in

Selection response can be as follows

$$R = h^2 \times SD \qquad SD = i \times \sigma_p$$

$$R = h^2 \times i \times \sigma_p$$

Where

- R = response
- i = intensity or standardized SD
- h^2 = heritability
- σ_P = phenotypic STD
- σ_g = genetic STD

We know that $h = \sigma_A / \sigma_p$

Or, $\sigma_A = h \times \sigma_p$

$$R = h \times i \times \sigma_A$$

Accuracy of Selection

- Accuracy in selection means **how closely the individual's real BV** agrees with what we think it is, when we do the selection
- More precisely, accuracy in selection means the **correlation b/n true BV and estimated BV**
- The accuracy of selection for a trait is tied directly to its h^2
- If h^2 is high, selection for a trait will permit an accurate sorting of BVs
- If h^2 is low, then many errors will be made when trying to select the best breeding animals from their own performance

Accuracy of Selection . . .

- **When the accuracy of selection on individuality is too low** to make reasonable genetic improvement, it can be enhanced by:

1. *Using additional measurements for the trait from the same individual*
2. *Using measurements of correlated traits*
3. *Using measurements on relatives (family selection), to predict BVs*

Intensity of selection (*the strength of selection to bring a genetic change*)

- The **intensity (i)** of selection is measured by the magnitude of the **selection differential (SD)**
- Does **not carry any unit** of measurement
- Intensity of selection attainable with truncation selection when specified fraction of a population are retained for breeding can be obtained from the table below

Per cent selected	P	i	Per cent selected	P	i
1	0.01	2.67	15	0.15	1.55
2	0.02	2.42	20	0.20	1.40
3	0.03	2.27	25	0.25	1.27
4	0.04	2.15	30	0.30	1.16
5	0.05	2.06	40	0.40	0.97
6	0.06	1.99	50	0.50	0.80
7	0.07	1.92	60	0.60	0.64
8	0.08	1.86	70	0.70	0.50
9	0.09	1.80	80	0.80	0.35
10	0.10	1.76	90	0.90	0.20

Just as for the **SD** in actual units, the **average intensity** is the average of the separate intensity values for the males & females [$\mathbf{i} = (\mathbf{i_m} + \mathbf{i_f}) / \mathbf{2}$].

Example 3: We wish to select for improved **weaning weight** in beef cattle.

$h^2 = 0.30$

herd mean = 400 lb

% saved (males) = 5%

% saved (females) = 30%

standard deviation = 40 lb

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selection intensity (i)

male 5% $i_m = 2.06$

female 30% $i_f = 1.16$

overall $i = (i_m + i_f) / 2 = (2.06 + 1.16) / 2 = \mathbf{1.61}$

Response

$$R = i \times h^2 \times \sigma_P = 1.61 \times .30 \times 40 = \mathbf{19.32} \text{ lb per gen.}$$

Generation interval (L)

- **L** is the average age of the parents when their offspring are born

L is found by simply calculating ***the average age of the parents when any offspring is born.***

Notice that **L** is ***not the average age when the 1st offspring is born.***

L is calculated separately for **males** & **female** and then averaged.

For example:

- ***Equal numbers of 2 & 3 year old bulls selected as parents*** L_{male} = (2+3)/2 = 2.5 years
- ***Equal numbers of 2, 3 & 4 year old cows selected as parents*** L_{female} = (2+3+4)/3 = 3.0 years
- $L_{average}$ = (2.5+3)/2 = 2.75 years.

Average generation interval

Species	Male	Female
Cattle	3 to 4	4.5 to 6
Sheep	2 to 3	4 to 4.5
Swine	1.5 to 2	1.5 to 3
Hoses	8 to 12	8 to 12
Chickens	1 to 1.5	1 to1.5

Variation

- Variation - differences among individuals within a population in measurable or observable traits
  - It is the **raw material** for herd/flock improvement
  - Potential for genetic improvement depends on the magnitude of variation among individuals within a population
  - Types of variation:
  - Continuous variation – observed in metric traits
  - Discontinuous variation – true for qualitative traits

Sources of variation

- The total phenotypic variation (VP) for traits in a population is due to:
  - **Genetics/heredity (VG)**
    - Additive (VA)
    - Dominance (VD)
    - Epistatic (VEp)
  - **Environment (VE)**
    - Temporary
    - Permanent
  - **Interaction between heredity and environment (VGE)**
    - VP = VG + VE + VGE

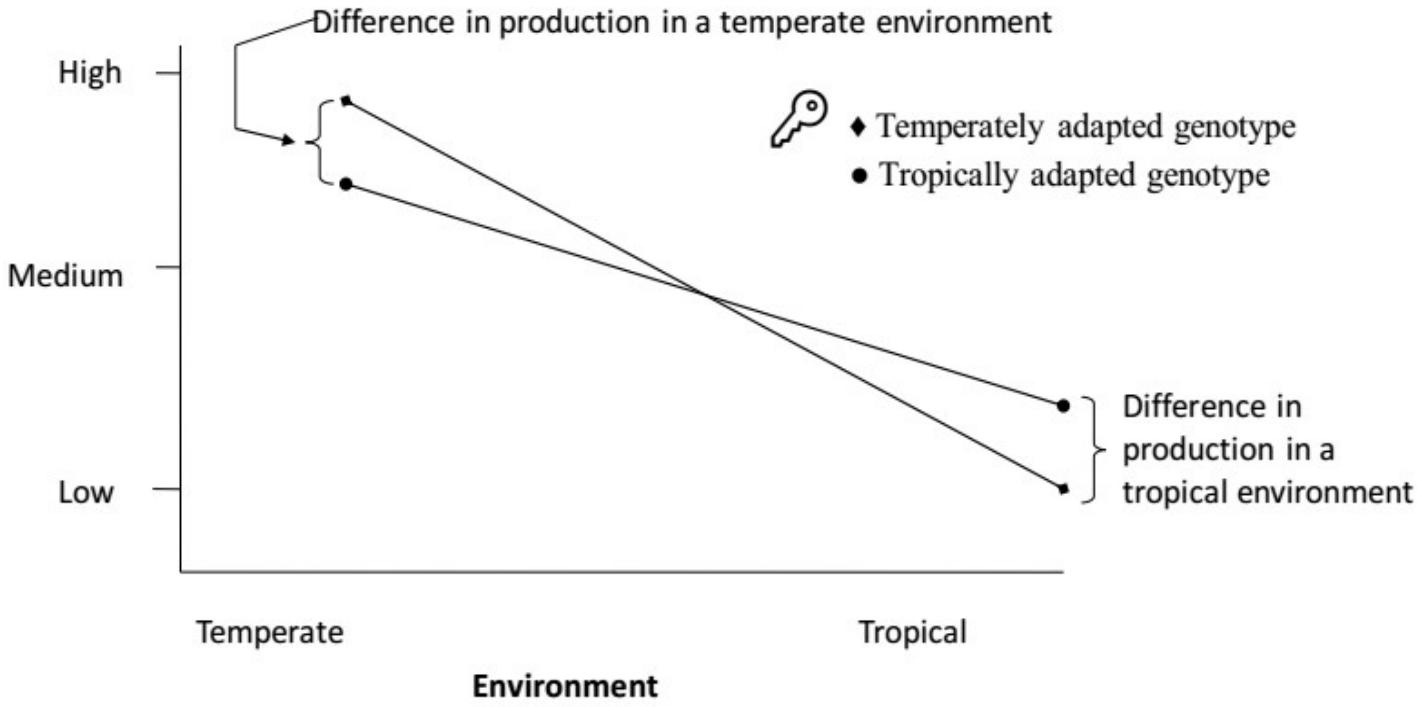
Properties of environmental variation

- Not transmitted through heredity
- Mask/overshadow variations due to heredity
- Proper environment is necessary for an individual to reach its genetic potential
- Uniform and **better environmental conditions** result in **rapid improvements** in the efficiency of livestock production

G x E interaction

- An interaction between **G × E** exists when the effect of genes is different in different environments
  - Then, the best animals in one environment for a given trait are not necessarily the best ones in another environment
  - To account for **G × E** interactions, the same trait measured under two different environments can be considered as two distinct traits
  - Genetic correlation between these trait-by environment combinations can be estimated
  - When the correlation is less than 1, animals do not rank equally in each environment

G x E interaction…



Important parameters and concepts

- Heritability – measures degree of resemblance between relatives (parent-offspring)
  - Correlation – measure of association between pairs of traits
    - Genetic correlation
    - Phenotypic correlation
  - Variance and standard deviation – measures of dispersion
    - Genetic variance/standard deviation
    - Phenotypic variance/standard deviation